

Chapter 14: Proportional Reasoning – 2 (Speed, Distance, and Time)

Worksheet 14: Speed, Distance, Time, and Applications

Learning Objective: Students will apply the relationship $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$ to solve problems involving uniform motion; work with unit conversions; solve problems on relative speed, average speed, and time-distance graphs.

Worked Example

A train 150 m long crosses a bridge 250 m long at 72 km/h. Find the time taken.

Solution: Total distance = $150 + 250 = 400$ m.

$$\text{Speed} = 72 \text{ km/h} = 72 \times \frac{1000}{3600} = 20 \text{ m/s}$$

$$\text{Time} = \frac{400}{20} = 20 \text{ seconds.}$$

Questions 1–5: Easy / Recall

Q1. A car travels 240 km in 4 hours. Find its speed.

Q2. How far does a cyclist travel at 15 km/h in 3.5 hours?

Q3. Convert 90 km/h into m/s.

Q4. Convert 15 m/s into km/h.

Q5. A train covers 420 km at 70 km/h. How long does the journey take?

Questions 6–10: Basic Application

- Q6.** Two cities are 450 km apart. Bus A leaves City 1 at 7:00 AM at 60 km/h; Bus B leaves City 2 at 8:00 AM at 90 km/h towards each other. At what time do they meet?
- Q7.** A man walks at 5 km/h for 2 hours, then cycles at 15 km/h for 1 hour. Find the average speed for the whole journey.
- Q8.** A train 200 m long passes a pole in 10 seconds. Find the train's speed in km/h.
- Q9.** If you walk at $\frac{3}{4}$ of your usual speed, you take 20 minutes extra to reach school. Find your usual time.
- Q10.** A plane flies 3000 km in 5 hours with wind assistance, but takes 6 hours against the wind. Find the speed of the plane in still air and the speed of the wind.

Questions 11–15: Practice and Skill Building

- Q11.** A thief steals a car and drives at 60 km/h. Police chase 10 minutes later at 90 km/h. How long before police catch the thief? How far from the starting point?
- Q12.** Two trains start from stations P and Q (500 km apart) towards each other at the same time. Speeds are 80 km/h and 70 km/h. When and where do they meet?

- Q13.** A boat goes 30 km upstream in 3 hours and 30 km downstream in 2 hours. Find the speed of the boat in still water and the speed of the current.
- Q14.** A car covers the first half of a journey at 40 km/h and the second half at 60 km/h. Find the average speed.
- Q15.** At what time should a train leave Station A to reach Station B (240 km away) by 2:00 PM if it runs at 80 km/h?

Questions 16–18: Higher Order Thinking

- Q16.** A train 300 m long overtakes another train 200 m long running in the same direction at 36 km/h. If the overtaking train runs at 54 km/h, how long does the overtaking take?
- Q17.** Person A starts from P towards Q at 8:00 AM at 5 km/h. Person B starts from Q towards P at 10:00 AM at 7 km/h. They are 44 km apart. At what time do they meet?
- Q18.** A student runs at 8 km/h and misses a bus by 2 minutes. If he had run at 10 km/h, he would have reached 2 minutes early. Find the distance to the bus stop.

Questions 19–20: Real-Life Applications

Q19. The Delhi–Mumbai Rajdhani Express covers 1384 km in about 17 hours. Find the average speed. If it stops for 30 minutes total, what is the speed during the actual running time?

Q20. Your school is 3 km from home. You walk at 4 km/h but want to reach by 8:00 AM for a 45-minute morning assembly. What is the latest time you can leave home? If you are 5 minutes late in leaving, how fast must you walk?

Reflection Question: Average speed is NOT the simple average of speeds for different stages of a journey (unless time is equal). Explain why, with an example.